Explicit Mathlib foundations

LLM-assisted specimen. Six source statements with separately checked Lean proofs.

Result 1 (addition commutes). [mathlib, add_comm]
Let x and y be real numbers. Then

$$x + y = y + x.$$

Result 2 (subtraction means addition of the negative). [mathlib, sub_eq_add_neg]
Let x and y be real numbers. Then

$$x - y = x + -y.$$

Adapted from the cited source.

Result 3 (intersection commutes). [mathlib, Set.inter_comm]
Let T be a type. Let A and B be subsets of T. Then

$$A \cap B = B \cap A.$$

Adapted from the cited source.

Result 4 (common part is contained). [mathlib, Set.inter_subset_left]
Let T be a type. Let A and B be subsets of T. Then

$$A \cap B \subseteq A.$$

Adapted from the cited source.

Result 5 (intersection membership). [mathlib, Set.mem_inter]
Let T be a type. Let x be an element of T. Let A and B be subsets of T. Assume $x \in A$ (`in A`). Assume $x \in B$ (`in B`). Then

$$x \in A \cap B.$$

Adapted from the cited source.

Result 6 (matrix product application). [mathlib, Matrix.mulVec_mulVec]
Let m, n and p be natural numbers. Let A be a real $m \times n$ matrix. Let B be a real $n \times p$ matrix. Let x be a real p-vector. Then

$$A\,B\,x = A\,(B\,x)\,.$$

Adapted from the cited source.